

Missionaries and Cannibals: Transition Rules

Rule 1: Moving from Left to Right

$$\begin{aligned} &L(M_1, C_1) \&\& R(M_2, C_2) \&\& (BL), \\ &\text{where } M, C \in \{0, 1, 2\}, \\ &M + C = 1 \parallel M + C = 2, \\ &(M_1 - M \geq C_1 - C) \parallel (M_1 - M = 0), \\ &(M_2 + M \geq C_2 + C) \parallel (M_2 + M = 0) \\ &\Rightarrow L(M_1 - M, C_1 - C) \&\& R(M_2 + M, C_2 + C) \&\& (BR) \end{aligned}$$

Rule 2: Moving from Right to Left

$$\begin{aligned} &L(M_1, C_1) \&\& R(M_2, C_2) \&\& (BR), \\ &\text{where } M, C \in \{0, 1, 2\}, \\ &M + C = 1 \parallel M + C = 2, \\ &(M_1 + M \geq C_1 + C) \parallel (M_1 + M = 0), \\ &(M_2 - M \geq C_2 - C) \parallel (M_2 - M = 0) \\ &\Rightarrow L(M_1 + M, C_1 + C) \&\& R(M_2 - M, C_2 - C) \&\& (BL) \end{aligned}$$